

Client-Side Scripting



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Bachelors of Science (Hons) (MQF Level 6)

Chapter 2 - VUE

What is Vue.js?

- ▶ Vue.js (VUE) is a JavaScript library for building web interfaces. Combined with some other tools it also becomes a framework.
- ▶ Both frameworks and libraries are code written by someone else that is used to help solve common problems.
- ▶ A framework is a 3rd party library that exposes utility functionalities and a set of rules on how to build your JS application.

What is Vue.js?

- ▶ The application is able to react to user input dynamically (REACTIVE); **look and feel of mobile apps.**
- ▶ HTML + CSS + JavaScript **running in the browser** of your user possibly talking to a server using Node or PHP.
- ▶ Theoretically we can build any kind of web application with just “vanilla” JavaScript without extra framework or library.

What is Vue.js?

- ▶ If we choose not to use any framework however:
- ▶ We will have to write **ALL CODE** and **LOGIC** and write everything from scratch; security issues; errors and bugs.
- ▶ And that is why we use a framework and focus on our business logic.

2 approaches to use VUE

1. VUE can be used to control **parts of HTML pages** or **entire pages**. Some pages are still rendered on and served by a backend server.
2. VUE can also be used to control the **entire frontend of a web application**. SPA approach. Server only sends one HTML page and then VUE takes over and controls the UI.

Building the first app with Vue.js

1. Use Visual Studio code which you can download for free and open the folder (01-first-app-with-just-jsFile) in Moodle. This is a **JavaScript example.**



Building the first app with Vue.js

1. Use Visual Studio code which you can download for free and open the 2nd folder in Moodle. This is a **VUE.js example**.

```
<body>
  <div id="app">
    <div>
      <label for="username">Username</label>
      <input type="text" id="username" v-model="enteredValue" />
      <button v-on:click="addUsername">Add Username</button>
    </div>
    <ul>
      <li v-for="username in usernames">{{ username }}</li>
    </ul>
  </div>
  <script src="https://unpkg.com/vue@next"></script>
  <script src="app.js"></script>
</body>
```

Building the first app with Vue.js

1. First we need to add/import Vue. There are different ways how to import VUE. Visit vuejs.org. For prototyping and learning we can use the following CDN:



The screenshot shows the Vue.js website. The header includes the Vue.js logo and navigation links: Docs, API Reference, Ecosystem, and Support. The left sidebar lists 'Essentials' with sub-links: Installation, Release Notes, Vue Devtools, and CDN. The main content area is titled 'CDN' and contains the text: 'For prototyping or learning purposes, you can use the latest version with:'. Below this, a code block shows the installation script: `<script src="https://unpkg.com/vue@next"></script>`.

Building the first app with Vue.js

1. A property named data must be added. The function must return an object. In this case it returns an array of usernames and the entered value. Initially both are empty. Vue is now aware of this data.
2. We then connect the entered value by using the directive **v-model** in the input.

```
<input type="text" id="username" v-model="enteredValue" />
```

```
Vue.createApp({  
  data() {  
    return {  
      usernames: [],  
      enteredValue: ''  
    };  
  },  
},
```

Building the first app with Vue.js

1. Now we want to add what is entered to the array usernames. Hence we create a method which is callable from inside the html using the directive **v-on:click**.

```
<button v-on:click="addUsername">Add Username</button>
```

```
methods: {  
  addUsername() {  
    this.usernames.push(this.enteredValue);  
    this.enteredValue = '';  
  }  
}
```

Building the first app with Vue.js

1. We then use the attribute **v-for** to replicate the element multiple times. We use the **double curly braces** to display all values in usernames array.

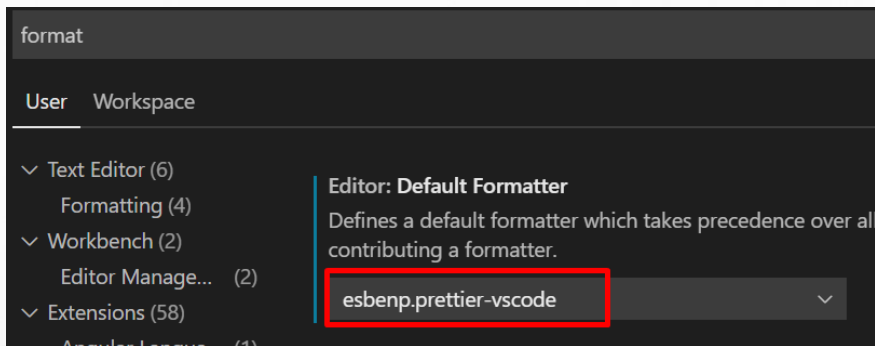
```
<li v-for="username in usernames">{{ username }}</li>
```

2. We then use mount to apply the Vue code on the app div element using the id selector #.

```
}).mount('#app');
```

Development environment

1. Visual Studio Code (install the **prettier** extension to format code and make it more readable).
2. Make sure it is set up by going to File – Preferences – Settings:



Course content

- ▶ **FOUNDATION:** Core syntax, Templates, Directives, Data, Methods, Computed Properties, Watchers.
- ▶ **INTERMEDIATE:** Components, Component communication, Forms, HTTP requests, Routing, Animation.
- ▶ **ADVANCED:** VUEX State Management, Authentication, Deploying and optimizing the VUE app, composition API, Re-using code.
- ▶ During this course its important you try things on your own and work out all practicals. If you get stuck use google and stackoverflow first and very often that will help you to solve the issue. If still stuck send your lecturer an email and do not hesitate to ask. Solutions will also be available in Moodle.

References

- Vue.js, 2020. *The Progressive JavaScript Framework*, s.l.: Vue.js.
- Schwarzmüller, M., 2020. *VUE JS - The Complete Guide*, s.l.: s.n.
- Jahr, A., 2020. *VUE Mastery*. [Online]
Available at: <https://www.vuemastery.com/courses/intro-to-vue-3/intro-to-vue3>
[Accessed 1 September 2020].

Any questions?

THANK YOU

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